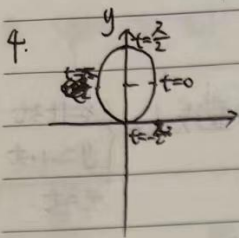


走师班补充作业5.

Record good times

chapter 11 #4, 7, 11, exam 3: 4.



$$S = 2 \int_0^{\frac{\pi}{2}} 2\pi x \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$= 4\pi \int_0^{\frac{\pi}{2}} 2\cos t \sqrt{(2\sin t)^2 + \cos^2 t} dt$$

$$= 8\pi \int_0^{\frac{\pi}{2}} \cos t \sqrt{3\sin^2 t + 1} dt$$

$$\underline{u = \sin t} \quad 8\pi \int_0^1 \sqrt{3u^2 + 1} du$$

$$\underline{u = \tan \theta} \quad \frac{8\pi}{\sqrt{3}} \int_0^{\frac{\pi}{3}} \sec^3 \theta d\theta$$

$$= \frac{8\pi}{\sqrt{3}} \cdot \frac{1}{2} \left(\sec \tan \theta + \ln |\sec \theta + \tan \theta| \right) \Big|_0^{\frac{\pi}{3}}$$

$$= 8\pi + \frac{4}{\sqrt{3}} \ln(2 + \sqrt{3})$$

$$\int \sec^3 \theta d\theta = \int \sec \theta d \tan \theta = \sec \theta \tan \theta - \int \tan^2 \theta \sec \theta d\theta = \sec \theta \tan \theta - \int \sec \theta d\theta + \int \sec \theta d\theta$$

$$\Rightarrow \int \sec^3 \theta d\theta = \frac{1}{2} \sec \theta \tan \theta + \frac{1}{2} \ln |\sec \theta + \tan \theta| + C$$

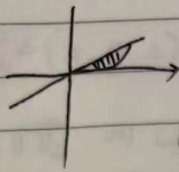
$$7. y' = \frac{dy/dt}{dx/dt} = \frac{2a \sin t \cos t}{-3a \cos^2 t \sin t} = -\frac{2}{3} \frac{1}{\cos t} = -\frac{2}{3} \sec t$$

$$y'' = \frac{dy'/dt}{dx/dt} = \frac{-\frac{2}{3} \sec t \tan t}{-3a \cos^2 t \sin t} = \frac{2}{9a \cos^3 t}$$

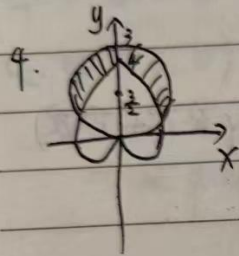
11. $r = \sin(4\theta)$

$\varphi\theta = 0 \Rightarrow \theta = 0, r = 0$

$\varphi\theta = \frac{\pi}{2} \Rightarrow \theta = \frac{\pi}{8}, r = 1$



$$A = 2 \int_0^{\frac{\pi}{8}} \frac{1}{2} r^2 d\theta = 2 \int_0^{\frac{\pi}{8}} \frac{1}{2} \sin^2(4\theta) d\theta = \int_0^{\frac{\pi}{8}} \frac{1 - \cos 8\theta}{2} d\theta = \frac{\pi}{16}$$



$$r = 3 \sin \theta = 1 \sin \theta \Rightarrow \theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$r = 3 \sin \theta \Rightarrow x^2 + (y - \frac{3}{2})^2 = \frac{9}{4}$$

$$A = 2 \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} \frac{1}{2} [9 \sin^2 \theta - (1 \sin \theta)^2] d\theta$$

$$= \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (8 \cdot \frac{1 - \cos 2\theta}{2} - 2 \sin^2 \theta) d\theta$$

$$= \pi.$$